$$> f(b,c,r) := \left(\frac{1}{2} - \cos\left(\frac{5}{6}\pi - c - r\right)\cdot\sin(c) - \frac{1}{2}\cdot\cos\left(\frac{2\cdot\pi}{3} + r\right) - \cos\left(\frac{\pi}{2} - b\right)\cdot\sin(b)\right)^2 + \left(\sin\left(\frac{5}{6}\pi - c - r\right)\cdot\sin(c) - \left(\frac{1}{2}\cdot\sin\left(\frac{2\cdot\pi}{3} + r\right) + \sin\left(\frac{\pi}{2} - b\right)\cdot\sin(b)\right)\right)^2$$

$$f := (b,c,r) \mapsto \left(\frac{1}{2} - \cos\left(\frac{5\cdot\pi}{6} - c - r\right)\cdot\sin(c) - \frac{\cos\left(\frac{2\cdot\pi}{3} + r\right)}{2} - \cos\left(-b + \frac{\pi}{2}\right)\cdot\sin(b)\right)^2 + \left(\sin\left(\frac{5\cdot\pi}{6} - c - r\right)\cdot\sin(c) - \frac{\sin\left(\frac{2\cdot\pi}{3} + r\right)}{2} - \sin\left(-b + \frac{\pi}{2}\right)\cdot\sin(b)\right)^2 \quad \textbf{(1)}$$

$$> g(b,c,r) := simplify\left(expand\left(f(b,c,0)^2 - f(b,c,r)\cdot f(b,c,-r)\right)\right)$$

$$g := (b,c,r) \mapsto simplify\left(expand\left(f(b,c,0)^2 - f(b,c,r)\cdot f(b,c,-r)\right)\right) \quad \textbf{(2)}$$

$$> collect(expand(g(b,c,r)), \cos(r))$$

$$-\frac{\cos(r)^2}{4} + \left(-\sqrt{3}\cos(c)^2\cos(b)\sin(b) - \cos(b)^2\cos(c)^2 + \frac{\cos(c)^2}{2} - \sqrt{3}\sin(c)\cos(c)\cos(b)^2 + \sin(c)\cos(c)\cos(b)\sin(b) + \frac{\sin(c)\sqrt{3}\cos(c)}{2} + \frac{\sqrt{3}\sin(b)\cos(b)}{2} + \frac{\cos(b)^2}{2} - \frac{1}{4}\right)\cos(r) + \sqrt{3}\cos(c)^2\cos(b)\sin(b) + \cos(b)^2\cos(c)^2 + \sqrt{3}\sin(c)\cos(c)\cos(b)^2 - \sin(c)\cos(c)\cos(b)\sin(b) - \frac{\sin(c)\sqrt{3}\cos(c)}{2} + \frac{\sin(c)^2}{2} - \frac{\sqrt{3}\sin(b)\cos(b)}{2} - \frac{\cos(b)^2}{2} \quad \textbf{(3)}$$

$$> simplify\left(\frac{1}{r^2}\left(\left(-\frac{1}{4}\right)\cdot(1-r^2) + \left(-\sqrt{3}\cos(c)^2\cos(b)\sin(b) - \cos(b)^2\cos(c)^2 + \frac{\cos(c)^2}{2} - \sqrt{3}\sin(c)\cos(c)\cos(b)^2 + \sin(c)\cos(c)\cos(b)\sin(b) + \frac{\sin(c)\sqrt{3}\cos(c)}{2} + \frac{\sqrt{3}\sin(b)\cos(b)}{2} + \frac{\cos(b)^2}{2} - \frac{1}{4}\right)\cdot\left(1 - \frac{r^2}{2}\right) + \sqrt{3}\cos(c)^2\cos(b)\sin(b) + \cos(b)^2\cos(c)^2 + \sqrt{3}\sin(c)\cos(c)\cos(b)^2 - \sin(c)\cos(c)\cos(b)\sin(b) - \frac{\sin(c)\sqrt{3}\cos(c)}{2} + \frac{\sin(c)^2}{2} - \frac{\sqrt{3}\sin(b)\cos(b)}{2} - \frac{\cos(b)^2}{2}\right)\right)$$

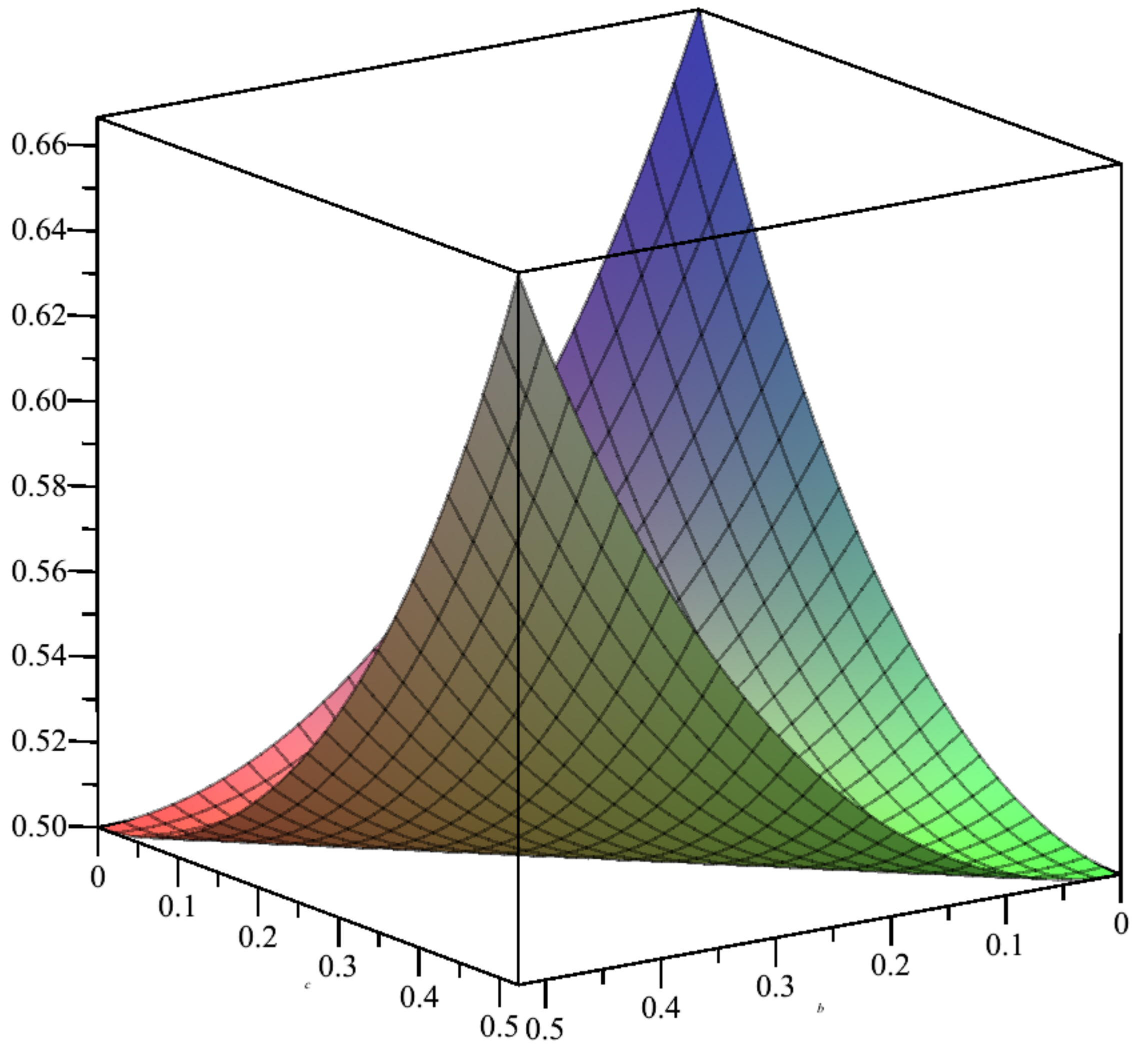
$$\frac{3}{8} + \frac{\left(2\sin(c)\cos(c)\cos(b)^2 + \sin(b)\left(2\cos(c)^2 - 1\right)\cos(b) - \sin(c)\cos(c)\right)\sqrt{3}}{4} + \frac{\left(2\cos(c)^2 - 1\right)\cos(b)^2}{4} - \frac{\sin(c)\cos(c)\cos(b)\sin(b)}{2} - \frac{\cos(c)^2}{4} \quad \textbf{(4)}$$

>

> $h(b, c) := \frac{3}{8}$
$$+ \frac{\left(2 \sin(c) \cos(c) \cos(b)^2 + \sin(b) \left(2 \cos(c)^2 - 1\right) \cos(b) - \sin(c) \cos(c)\right) \sqrt{3}}{4}$$
$$+ \frac{\left(2 \cos(c)^2 - 1\right) \cos(b)^2}{4} - \frac{\sin(c) \cos(c) \cos(b) \sin(b)}{2} - \frac{\cos(c)^2}{4}$$

$$h := (b, c) \mapsto \frac{3}{8} + \frac{\left(2 \cdot \sin(c) \cdot \cos(c) \cdot \cos(b)^2 + \sin(b) \cdot \left(2 \cdot \cos(c)^2 - 1\right) \cdot \cos(b) - \sin(c) \cdot \cos(c)\right) \cdot \sqrt{3}}{4} + \frac{\left(2 \cdot \cos(c)^2 - 1\right) \cdot \cos(b)^2}{4} - \frac{\sin(c) \cdot \cos(c) \cdot \cos(b) \cdot \sin(b)}{2} - \frac{\cos(c)^2}{4} \tag{5}$$

> $plot3d\left(\frac{h(b, c)}{f(c, b, 0)^2}, b = 0 .. \frac{3.14}{6}, c = 0 .. \frac{3.14}{6}\right)$



>

> $int\left(int\left(\frac{h(b,c)}{f(c,b,0)^2}, b = 0 .. \frac{3.1415926535}{6}, numeric = true\right), c = 0 .. \frac{3.1415926535}{6}, numeric = true\right)$

$$0.1438410363 \tag{6}$$

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